

# FR-HHT Locking & Chirp — Interim Update

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## Executive summary

HHT IMF2 locking: baseline axis  $\ln(1+z)$  ( $\alpha=1.0$ ) outperforms FR-style axis ( $\alpha=0.5$ ) on  $|r|$ ; p-values tied at 0.00498 for the tested pairs.

FR SN+BAO (spline  $\Lambda$ ): allowing  $c(z)$  in BAO ('c\_of\_z') gives a modest  $\chi^2$  improvement vs constant-c in BAO ('c0').

Piecewise- $\Lambda$  'discrete events':  $\Delta z \rightarrow 1e-4$  reproduces the continuous result within rounding; coarse  $\Delta z=0.015$  is slightly worse (as expected).

OU  $\Lambda(z)$  sensitivity: results stable from  $\text{amp}=1e-120 \dots 1e-70$ ; degrade clearly by  $\text{amp} \approx 1e-2$ .  
Corr-length sweep (0.05–0.60) showed no material change.

PBH  $\Lambda \leftrightarrow A_{\text{acc}}$  coupling  $\gamma \in \{0, 0.02, 0.05, 0.10\}$ : no change in best-fit or  $\chi^2$  with SN-only — suggests the tested  $\gamma$  range is too small or SN-only lacks constraining power.

Soft-link (energy–luminosity) with flat FR and  $\log_{10} p_z$  scalings: initial  $\kappa = \{-1.0, -0.75\}$ ,  $\sigma=0.10$  runs converge to identical solutions; likely penalty not binding at this  $\sigma$ .

## HHT IMF2 locking ( $\alpha=0.5$ vs $\alpha=1.0$ )

Setup: ceemdan, IMF=2, packs=bao\_hht\_2bin\_z0p7\_longbest, seeds={3,7,11}, ns=200.

Summary table: mean $|r| \rightarrow \alpha=0.5$ : 0.8790809752,  $\alpha=1.0$ : 0.9093863675; median p both 0.0049751244; win rate ( $\alpha=0.5$ ) = 0/3.

## FR (spline $\Lambda$ ) with SN+BAO — BAO c-mode comparison

c0:  $\chi^2=2017.2954$ ,  $\chi^2/\text{dof}=1.18247$ , AIC=2031.2954, BIC=2069.4174,  $H_0=59.576$ ,  $\Omega_m=0.3435$ ,  $\Omega_k=0.00779$ ,  $w_{\text{eff}} \approx -0.912$ .

c\_of\_z:  $\chi^2=2014.5488$ ,  $\chi^2/\text{dof}=1.18086$ , AIC=2028.5488, BIC=2066.6709,  $H_0=60.579$ ,  $\Omega_m=0.1333$ ,  $\Omega_k=0.25117$ ,  $w_{\text{eff}} \approx -1.080$ .

'c\_of\_z' gains  $\sim 2.75$  in  $\chi^2$  vs 'c0' with these settings; parameters shift toward higher  $|\Omega_k|$  and more negative  $w_{\text{eff}}$ .

## Piecewise- $\Lambda$ : continuous vs discrete-events

continuous:  $\chi^2=1975.2424$ , AIC=1989.2424 (baseline)

discrete  $\Delta z=0.015$ :  $\chi^2=1980.4284$ , AIC=1994.4284 (coarser events worse)

discrete  $\Delta z=1e-4$ :  $\chi^2=1975.2100$ , AIC=1989.2100 (matches continuous within rounding)

## OU $\Lambda(z)$ sensitivity

Correlation-length sweep (amp=1e-120):

corr<sub>z</sub> ∈ {0.05, 0.15, 0.30, 0.60} → identical best-fit &  $\chi^2=1974.8730$  (SN-only).

Amplitude sweep at fixed corr<sub>z</sub>=0.30 and fixed parameters ('sampler=none'): 1e-120...1e-70 unchanged; 1e-2 degrades ( $\chi^2 \approx 2089.88$ ).

Probe MCMC at amp=1e-2 (short chain):  $\chi^2 \approx 2020.38$ .

## PBH $\Lambda \leftrightarrow A_{\text{acc}}$ coupling ( $\gamma$ ) — SN-only

$\gamma \in \{0.00, 0.02, 0.05, 0.10\} \rightarrow \chi^2=1974.5117$  for all; no effect across this range for SN-only.

Action: include BAO/Planck and/or widen  $\gamma$  grid.

## Soft-link experiments (flat FR, log1pz scalings)

$\kappa=-1.0, \sigma=0.10 \rightarrow \chi^2=2005.9505, \text{AIC}=2017.9505$ ;

$\kappa=-0.75, \sigma=0.10 \rightarrow \chi^2=2005.9505, \text{AIC}=2017.9505$ ;

Identical outcomes suggest the soft constraint isn't binding at  $\sigma=0.10$ .

Action: repeat for  $\sigma \in \{0.05, 0.02, 0.01\}$ .

## Data & environment notes

Planck prior: 'planck\_distance\_prior.json' missing in /Users/boyde/Downloads/ — add file to enable --planck and --planck-c-mode tests.

emcee: use nwalkers  $\geq 14$  for 7 parameters; corner is optional.

HHT  $\alpha$ -compare DOCX needed python-docx (installed).

## Recommended next steps

HHT:  $\alpha \in \{0.6 \dots 1.4\}$  (step 0.1) for IMF2/IMF3; keep surrogate trials; report  $|r|$  and p-value distributions per pad/method.

BAO/Planck: once Planck JSON present, compare  $c_0$  vs  $c_{\text{of}_z}$  under Planck distance priors; inspect  $\Omega_k$  ion

Discrete events: quantify convergence vs  $\Delta z$  ladder ( $1e-3$ ,  $5e-4$ ,  $1e-4$ ) with AIC/BIC.

OU  $\Lambda(z)$ : place informative priors on  $\text{corr}_z$  and amplitude; test rejection thresholds for large amplitudes ( $\geq 1e-4$ ).

PBH coupling: extend  $\gamma$  grid and include BAO/Planck; track posteriors of  $A_{\text{acc}}$ ,  $n_{\text{acc}}$ , and  $\gamma$ .

Soft-link: sweep  $\kappa \in [-1.5, +1.5]$  with  $\sigma \in \{0.05, 0.02, 0.01\}$ ; ensure  $n_{\text{steps}} \geq 2500$ .

## Reproduct: Command Snippets

### # FR + SN+BAO (spline $\Lambda$ ): $c_{\text{of}_z}$

```
%run '/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \  
--model FR --data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \  
--cov '/Users/boyde/Downloads/Pantheon+SH0ES_STAT+SYS.cov' \  
--bao '/Users/boyde/Downloads/bao_long.csv' --rd 147.09 \  
--sampler emcee --nwalkers 24 --nsteps 800 --nburn 200 --nint 220 --seed 73 \  
--fluct-Lambda --Lambda-model spline --Lambda-amp 1.0e-120 --Lambda-knots 7 \  
--bao-c-mode c_of_z --out 'plamb_runs/SNBAO_baocmode_c_of_z'
```

### # Piecewise- $\Lambda$ : $\Delta z \rightarrow 1e-4$ matches continuous

```
%run /Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py \  
--model FR --data /Users/boyde/Downloads/Pantheon+SH0ES.dat \  
--cov /Users/boyde/Downloads/Pantheon+SH0ES_STAT+SYS.cov \  
--sampler none --nint 400 \  
--H0 82.5742 --Om 0.68207 --Ol 0.96443 --A_acc -0.25624 --n_acc -5.84476 \  
--gamma_c -0.02980 --epsilon_M 0.38324 \  
--fluct-Lambda --Lambda-model piecewise --Lambda-amp 1e-120 --Lambda-knots 8 \  
--discrete-events --event-size 0.0001 \  
--out 'plamb_runs/SN_piecewise_disc_at_cont_DZ1e-4'
```